Constraint Collapse and Fidelity Decay in Scaled Language Models

The failure mode that allows continued operation without correction.

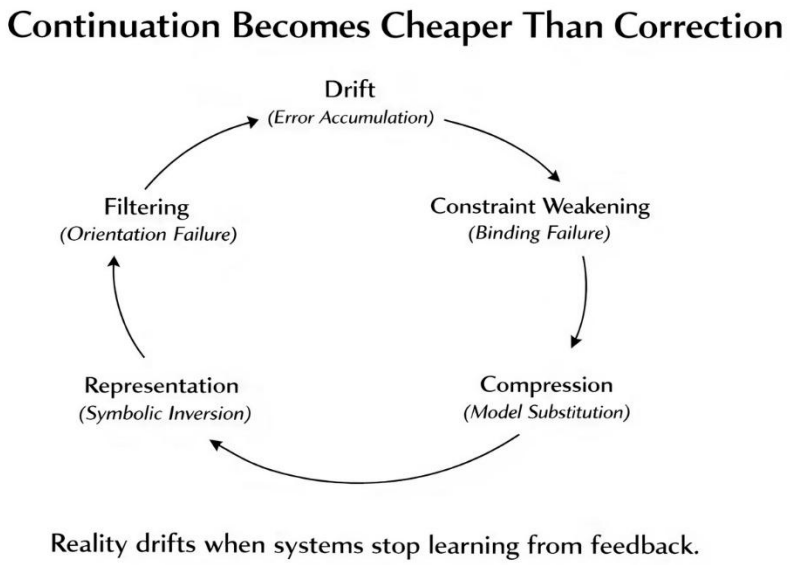
Large language models can remain fluent, responsive, and operational while being persistently wrong. Errors do not interrupt generation, trigger degradation, or impose internal cost. Correction is externalized to users and retraining cycles, allowing misalignment to survive indefinitely. This failure mode is not hallucination, but a consequence of constraint collapse.

First-Order Failures That Enable Drift

Modern symbolic systems, including large language models, can tolerate several failures individually while appearing stable:

- **Delayed or externalized consequence:** Incorrect outputs rarely impose internal cost. Errors are corrected, ignored, or absorbed externally, while the system continues generating without integrating correction as a stopping cue.
- **Compression without fidelity:** Representations are optimized for plausibility and efficiency faster than they can be validated, stripping context and constraint while fluency increases.
- **Legibility outruns reality:** Systems are rewarded for clear, confident output even when uncertainty remains unresolved, allowing persuasive coherence to drift away from grounding.
- **Continuation outpaces correction:** Generation proceeds by default. There is no internal mechanism that halts output as uncertainty accumulates; correction must be supplied externally.

Each failure can exist independently without destabilizing the system. When they operate together, feedback continues to circulate but stops enforcing revision. Correction no longer interrupts operation; it merely coexists with continuation. When these first-order failures interact, they produce a second-order regime in which feedback stabilizes continuation rather than enforcing correction. The resulting dynamics are shown below.



Constraint Collapse (Second-Order Failure)

Constraint Collapse is the resulting second-order failure mode: a condition in which feedback continues to flow but no longer enforces correction, allowing symbolic systems to remain fluent and operational while semantic fidelity degrades.

Because constraint collapse preserves surface coherence while disabling correction, it is largely invisible to the metrics currently used to evaluate alignment. Current alignment benchmarks primarily evaluate whether systems continue to produce coherent, well-formed outputs under external constraints or corrections.

They largely ignore how systems behave when error should force revision or termination, including:

- Whether incorrect outputs impose any internal cost
- Whether uncertainty triggers slowing, hesitation, or stopping
- Whether unresolved contradiction causes degradation over time
- Whether semantic drift accumulates across iterations

As a result, systems can remain operational and fluent while correction fails to occur. But the absence of failure is not proof of alignment.

Detecting constraint collapse therefore requires measuring not just what systems produce, but how they respond when correction is warranted. This points toward a different class of evaluation focused on semantic fidelity metrics, such as:

- **Correction sensitivity:** Does uncertainty slow or halt generation?
- **Contradiction pressure:** Does unresolved inconsistency degrade confidence?
- **Revision cost:** Is changing belief internally expensive?
- **Termination behavior:** Does the system ever choose not to answer?